

CSC 225 FALL 2024
ALGORITHMS AND DATA STRUCTURES I
ASSIGNMENT 1 - WRITTEN
UNIVERSITY OF VICTORIA

1. [4 marks] For each of the following determine the number of arrangements of the given letters given the specific restrictions.
 - a) “ALGORITHMS” with no restrictions.
 - b) “DATASTRUCTURES” with no restrictions.
 - c) “ALGORITHMS” where all the vowels are adjacent.
 - d) “DATASTRUCTURES” where all the letters in “DATA” are not adjacent. That is, at least one of the letters in “DATA” is separated from the rest by at least one letter.
2. [4 marks] Suppose that you draw five cards from a standard deck of 52.
 - a) How many ways can you draw exactly 2 spades?
 - b) How many ways can you draw at most 2 spades?
 - c) How many ways can you draw exactly 2 spades and 2 hearts?
 - d) How many ways can you draw exactly 1 spade, 2 hearts and 2 jacks, where there may be overlap (that is, 1 jack may be a spade, and/or 1 jack may be a diamond, etc.)?
3. [4 marks] Show that for all integers n, k , such that $0 < k < n$, then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

4. [4 marks] Determine the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 32$, where
 - a) $x_i \geq 0, 1 \leq i \leq 4$
 - b) $x_1, x_2 \geq 2, x_3, x_4 \geq 1$
 - c) $x_i \geq -1, 1 \leq i \leq 4$
 - d) $x_i \geq 0, 1 \leq i \leq 3$, and $2 \leq x_4 \leq 7$
5. [4 marks] Let A be a subset of $\{1, 2, 3, \dots, 25\}$ where $|A| = 9$. For any subset B of A let s_B denote the sum of the elements in B . Prove that there are distinct subsets C, D of A such that $|C| = |D| = 5$ and $s_C = s_D$.

Question 1

a) ALGORITHMS [no restrictions]

$$\begin{aligned}\text{number of arrangements} &= 10! \\ &= 3,628,800\end{aligned}$$

b) DATASTRUCTURES [no restrictions]

repeated letters:

A, T, S, R, U

$$\text{number of arrangements} = \frac{14!}{2! \times 3! \times 2! \times 2! \times 2!} = 908107200$$

c) ALGORITHMS
[vowels are adjacent]

Arrangements of vowels: $3!$

Arrangements of consonants: $8!$

$$\text{total} = 40320 \times 6 = 241,920$$

Question 2

a) A deck has 13 spades.

We need to choose exactly 2 from the 13

→ no of ways to choose 2 spades from 13:

$$\binom{13}{2} = \frac{13 \times 12}{2 \times 1} = 78$$

→ no of ways to choose 3 non-spades from 39 cards

$$\binom{39}{3} = \frac{39 \times 38 \times 37}{3 \times 2 \times 1} = 9139$$

$$\text{Total no. of ways} = 9139 \times 78 = 712,842$$

Question 2

b) no. of ways to choose 0 spades: $\binom{39}{5}$

no. of ways to choose 1 spade: $\binom{39}{4} \times \binom{13}{1}$

no. of ways to choose 2 spades: $\binom{13}{2} \times \binom{39}{3}$

$$\text{total} = \binom{39}{5} + \binom{13}{1} \binom{39}{4} + \binom{13}{2} \binom{39}{3} = 2,357,862$$

c) no. of ways to choose exactly 2 spades: $\binom{13}{2}$

no. of ways to choose exactly 2 hearts: $\binom{13}{2}$

no. of ways to choose exactly 1 card
that is neither a spade nor a heart: $\binom{26}{1}$

$$\begin{aligned} \text{total no. of ways: } & \binom{13}{2} \times \binom{13}{2} \times \binom{26}{1} \\ & = 158,184 \end{aligned}$$

Question 3

for all integers n, k st $0 < k < n$

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

given that $k < n$

$$\binom{n-1}{k-1} + \binom{n-1}{k}$$

$$= \frac{(n-1)!}{(k-1)! (n-k)!} + \frac{(n-1)!}{k! (n-k-1)!}$$

$$= (n-1)! \left(\frac{k}{k! (n-k)!} + \frac{n-k}{k! (n-k)!} \right)$$

$$= (n-1)! \left(\frac{n}{k! (n-k)!} \right)$$

$$= \frac{n!}{k! (n-k)!} = \binom{n}{k}$$

Hence for all integers n, k st $0 < k < n$

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Question 4

Determine the number of integer solutions of

$$x_1 + x_2 + x_3 + x_4 = 32, \text{ where}$$

a) $x_i \geq 0, 1 \leq i \leq 4$

$$\text{Formula} = \binom{n+k-1}{k-1}$$

$$= \binom{32+4-1}{4-1} = \binom{35}{3}$$

$$= 6545$$

b) $x_1, x_2 \geq 2, x_3, x_4 \geq 1$

let

$$y_1 = x_1 - 2 \quad y_3 = x_3 - 1$$

$$y_2 = x_2 - 2 \quad y_4 = x_4 - 1$$

$$(y_1 + 2) + (y_2 + 2) + (y_3 + 1) + (y_4 + 1) = 32$$

$$y_1 + y_2 + y_3 + y_4 + 6 = 32$$

$$y_1 + y_2 + y_3 + y_4 = 26$$

$$n = 26, k = 4$$

$$\binom{n+k-1}{k-1} = \binom{26+4-1}{4-1} = \binom{29}{3}$$

$$= 3654$$

$$c) x_i \geq -1, 1 \leq i \leq 4$$

$$y_1 = x_1 + 1$$

$$y_2 = x_2 + 1$$

$$y_3 = x_3 + 1$$

$$y_4 = x_4 + 1$$

$$(y_1 - 1) + (y_2 - 1) + (y_3 - 1) + (y_4 - 1) = 32$$

$$y_1 + y_2 + y_3 + y_4 = 36$$

$$\binom{n+k-1}{k-1} = \binom{36+4-1}{4-1} = \binom{39}{3}$$

$$= \boxed{9139}$$

$$d) x_i \geq 0, 1 \leq i \leq 3, \text{ and } 2 \leq x_4 \leq 7$$

$$\text{Case i: when } x_4 = 2$$

$$x_1 + x_2 + x_3 = 30$$

$$\text{Case ii: when } x_4 = 3$$

$$x_1 + x_2 + x_3 = 29$$

$$\text{Case iii: when } x_4 = 4$$

$$x_1 + x_2 + x_3 = 28$$

$$\text{Case iv: when } x_4 = 5$$

$$x_1 + x_2 + x_3 = 27$$

$$\text{Case v: when } x_4 = 6$$

$$x_1 + x_2 + x_3 = 26$$

$$\text{Case vi: when } x_4 = 7$$

$$x_1 + x_2 + x_3 = 25$$

$$\begin{aligned} \text{Total} &= \binom{32}{2} + \binom{31}{2} + \binom{30}{2} + \binom{29}{2} + \binom{28}{2} + \binom{27}{2} \\ &= 2531 \end{aligned}$$

Question 5

A is a subset of $\{1, 2, 3, \dots, 25\}$ where $|A| = 9$

s_B = sum of elements in B

B is a subset of A .

Prove that there are distinct subsets C, D of A such that $|C| = |D| = 5$ and $s_C = s_D$

→ Total no. of subsets:

Since A has 9 elements.

$$= 2^9 = \underline{512}$$

→ sum range of subsets:

maximum possible sum is =

$$17 + 18 + 19 + 20 + 21 + \dots + 25 = 189$$

→ Pigeonhole Principle:

There are 512 subsets of A
and the sum of elements in each
subset must be an integer between
0 and 189 (inclusive).

It gives us 190 possible distinct sums.

By pigeonhole principle there exists distinct
subsets C and D such that $s_C = s_D$